Status	Finished
Started	Monday, 12 October 2026, 6:54 PM
Completed	Monday, 12 October 2026, 6:57 PM
Duration	3 mins 25 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is sorted ✓
c. When the list is very small
d. When the list is unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The search is considered unsuccessful ✓
c. The last element is returned
d. An error is raised

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. There must be a mechanism to access middle element directly
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Sorting the list before searching
c. Checking each element sequentially ✓
d. Dividing the list in half repeatedly

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 94
c. 89 and 94
d. 90 and 99 ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **7**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Binary search ✓
c. Hash search
d. Sequential search

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓
d. The list must be sorted

Question **9**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Finding
c. Searching ✓
d. Mining

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search requires the list to be sorted.
c. Linear search divides the list into halves.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By dividing the list into halves
c. By using a hash function
d. By sorting the list first

Question **12**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search continues in the right sublist
c. The list is sorted
d. The search ends successfully ✓

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. Item is the last element in the array or item is not there at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is the last element in the array

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Insertion

d. Selection

Question **15**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists

b. It always starts from the beginning of the list

c. It compares elements sequentially

d. It can be applied only if the list is sorted ✓